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Domain cloning for a particle-in-cell (PIC) code on a cluster of symmetric-multiprocessor (SMP) computers

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Abstract

In 2000, a new approach to parallelization, domain cloning, was presented by [C.C. Kim, S.E. Parker, Massively parallel three-dimensional toroidal gyrokinetic flux-tube turbulence simulation, Journal of Computational Physics 161 (2000) 589–604]. Domain cloning is an additional layer of parallelization, a supplement to one-dimensional domain decomposition which gives the opportunity to optimize the scaling properties of certain particle-in-cell codes. A further adaptation to a cluster of symmetric-multiprocessor computers is presented in this work. Hence, very good scaling properties up to 2048 processors could be achieved for a particle-in-cell code surpassing the Tflop/s performance threshold.
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1. Introduction

The nonlinear particle-in-cell (PIC) code TORB [1] simulates a plasma in a cylindrical θ-pinch configuration which is a first approximation of the full 3D geometry of the stellarator Wendelstein 7-X (W7-X) [2]. The code solves the coupled system of gyrokinetic equations for the ions, in the electrostatic approximation, and the quasi-neutrality equation, assuming adiabatically responding electrons.

PIC codes in general have been used in the plasma physics community for several decades, and there are a number of textbooks available on the subject [3,4]. A skeleton PIC code for parallel computers is given in Ref. [5]. Applied to the TORB code, this gives the following main structure: the ions interact with the electrostatic field created by their own charge density. Thus, the code has two distinct parts and data structures. In one part, particles interact with the electrostatic field, this interaction is described by the equation of motions. The corresponding set of coupled differential equations is integrated in time using an explicit fourth-order Runge–Kutta method. In the other part of the code, the fields created by the particles must be found. This is done by solving the quasi-neutrality equation. Its source term is the gyro-averaged ion density which

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is calculated from the particle positions in a process called charge-assignment. Finite elements are used to represent both the electrostatic potential and the particle shape function in the charge-assignment. The finite elements used are a tensor product of cubic B-splines (for an introduction see Ref. [6]) and guarantee in principle the physical conservation laws of particle number and energy conservation.

In general, global nonlinear PIC simulations of turbulent field structures are very demanding with respect to numerical methods and computational effort. This is due to the fact that the five-dimensional distribution function of the ion guiding centers has to be sampled. In addition, the unavoidable statistical noise at the charge-assignment has to be reduced to a minimum. Thus, the TORB code uses two statistical noise reduction techniques. The first one is well-known in the plasma physics community as the δf method [7]. It can be identified as a control variates method for Monte Carlo simulations which tries to reduce the variance of physical quantities, e.g. particle density, by replacing as much of the Monte Carlo estimate as possible by analytic calculations that are free of statistical error. The TORB code uses it for the discretization of the five-dimensional ion guiding center distribution function f so that particle resolution is not wasted in representing the background Maxwellian distribution function f_0. The second one is an importance sampling technique introduced in Ref. [1] for nonlinear δf PIC simulations. The initial position of the Monte Carlo particles in phase space is optimized by a pilot run which provides information about the $|\delta f|$ function evolving at late times of the simulation.

Despite of the two variance reduction techniques a minimal numerical simulation still consists of $\approx$100 million Monte Carlo particles which makes it necessary to run the TORB code on massively parallel computers with good scaling properties.

Section 2 gives a brief introduction to domain decomposition and particle decomposition techniques and their advantages and disadvantages. The hybrid method of both, domain cloning, is introduced in Section 3. Results produced with the domain cloning concept on a cluster of symmetric-multiprocessor (SMP) computers are presented in Section 4. Finally a summary is given in Section 5.

2. Domain decomposition vs. particle decomposition

The essential idea of domain decomposition is that different portions of the physical domain and of the corresponding electrostatic field grid are assigned to different processing elements (PEs), together with the Monte Carlo particles that reside on them. As particles move from one region to another, they are communicated to the PE which is associated with the new region. Thus, the number of particles resident on each PE is not a constant of time which could cause load imbalance. During the calculation of the gyro-averaged electric field at the particle position, field information of neighboring regions of space must be accessed, so that extra guard cells are kept in each PE's memory. Special routines take care that the particles are passed to the correct PE when leaving a domain and that the guard cells are updated after a calculation of the electrostatic potential.

The particle sorting and communicating algorithm was first developed by V.K. Decyk as part of the "Plib" parallel PIC simulation library [5], and further developed by Tran [8]. It involves an initialization routine and a move routine described in Ref. [9]. The initialization routine allocates an array for the particles that will be passed to other subdomains (*send*), an array to index the particles to be passed (*isend*), and an array to index the holes in the particle array (*ihole*). The move routine determines which particles do not belong to the subdomain (and simultaneously which subdomain they do belong to; usually the subdomains of neighboring PEs) and records their indices in *isend* and *ihole*, then fills and sorts (by subdomain) the send array. The appropriate portion of the *send* array is passed to the appropriate subdomain. The particle array is then filled, first filling the holes using *ihole* then appending to the end of the array. An additional modification by M. Borchardt makes it possible to send the full information belonging to one particle in one sweep rather than to pass only one attribute of the particle at a time.

The concept of one-dimensional domain decomposition has a major disadvantage when the number of PEs is increased to share the particle workload among a larger number of PEs. The number of subdomains scales with the number of PEs so that the spatial extent in the direction of parallelization becomes smaller and smaller. The average time spent by the particles in a subdomain decreases causing an increased communication of particles between subdomains which are no longer exclusively neighboring. In addition, the grid resolution in the direction of parallelization becomes larger and larger and is decoupled from the typical spatial scale of the simulated field structures which should limit the grid resolution.

Another concept called particle decomposition [10] uses the following strategy: the whole spatial grid is assigned to each PE, which however takes care only of a subset of the particle population. Partial contributions to the gyro-averaged ion density, which is required to update the electrostatic field, are communicated among PEs and summed together. Thus, this approach presents, compared with the domain decomposition concept, the advantage that it has an intrinsic load balancing and avoids particle communication. Only if the number of communicated particles is much larger than the number of B-splines representing the electrostatical potential it is advantageous to broadcast the electrostatic field data in a global sum rather than communicating the particles over subdomain boundaries. This is paid for by a higher memory consumption per PE, because each PE has to store the full field data, and an expensive all-reduce communication (global sum) over the PEs has to be carried out to update the field data stored on each PE.

Nonlinear simulations as within the TORB code usually require a very small time step Δt which restricts even the fastest particles to travel only a short distance in the direction of parallelization. Consequently in case of domain decomposition, only a small fraction of the particles leaves the subdomains, and particles are only communicated between neighboring PEs. This makes particle communication local and inexpensive compared to global field data communication associated with particle decomposition which becomes expensive with increasing number of PEs.

3. Domain cloning

The domain cloning concept is a combination of particle decomposition and domain decomposition including both of them as limiting cases. According to this, which was introduced in Ref. [9], the simulation domain is cloned, i.e. multiple copies of the same domain are made. The PEs are divided into a number of groups equal to the number of clones. Each group of PEs is assigned to one of the domain clones. Each domain clone is loaded with its own set of particles and a one-dimensional decomposition is performed such that each PE has a corresponding subdomain clone. The concept of domain cloning is illustrated in Fig. 1 as an example of two clones of a periodic cylindrical simulation domain. The over-all number of 16 PEs is divided into two groups of eight PEs, and each group is assigned to one of the domain clones. On each clone a one-dimensional domain decomposition is performed along the cylinder axis (the direction of fast motion of the ions parallel to the B-field). Thus, the subdomain clones for which the global sum has to be performed lie opposite to each other as illustrated in Fig. 1.

As each clone has a copy of the simulation domain, it has to store the full field data. However, this is much less restrictive compared to the particle decomposition where each PE has to store the full field data by itself. If

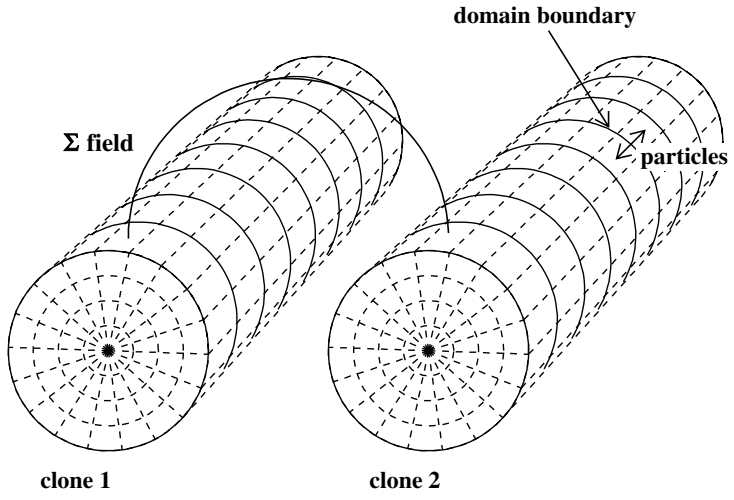


Fig. 1. A two-fold domain clone of the simulation domain. Particle communication occurs inside the clones; field data are summed over subdomain clones.

one associates with each PE a certain amount of memory, the more PEs assigned to each clone, the larger the total amount of memory which can be used to store the field data. Thus, the minimal number of PEs per clone could be limited in principle by the memory necessary to store the filed data, especially when the number of particles per grid cell is small. But in practice, it has been shown that this is not a major restriction as the typical number of PEs per clone is quite large, e.g. 32 PEs.

The realization of the parallelization can be easily performed with the Message Passing Interface (MPI). Two communicators are defined as follows: one for the communication inside the domain clone used for the particle communication and the update of the guard cells, the other for the global sum over the field data across the subdomain clones. Each domain clone has identical field data, but retains its own set of particles. Hence, particle communication is reduced as each cloned domain has subdomains which span a larger portion of the domain as compared to the one-dimensional decomposition. In addition, there is no longer direct scaling between the grid resolution in the direction of parallelization and the number of PEs.

A one-dimensional domain decomposition is already implemented in the original version of the TORB code. As seen above, the domain cloning algorithm can be easily developed from one-dimensional domain decomposition. So only minor changes to the code had to be done to introduce domain cloning through the concept of the two MPI communicators explained. In doing so, one achieved an additional degree of freedom to mix the domain- and particle decomposition concepts. This degree of freedom has been used to adapt domain cloning to the special hardware constraints of a cluster of SMP computers to significantly optimize the scaling properties of the PIC code TORB. Today's clusters of SMP computers are usually characterized by a hierarchical structure concerning data access: The bandwidth inside the shared memory of a single compute node is higher than the bandwidth via the network between nodes. If now the compute nodes are identified with the clones, data intensive particle communication is restricted to the shared memory of a node and is fast compared to less intensive field communication between nodes which has to pass the slower network.

With the increasing number of installed clusters of SMP computers in the past years, the hybrid MPI/OpenMP paradigm became the software trend on these architectures. It is elegant in concept and fits perfectly the architecture: using MPI across nodes and OpenMP within nodes. Thus, in theory, a good usage of the resources of the SMP computer e.g. memory and bandwidth is guaranteed, while extra communication overhead due to MPI within the node is avoided. However, for the TORB code the standard MPI use for both communication across the nodes and handling of concurrent memory access within the node has not changed yet. The author's experience with other parallel codes shows that at least the IBM Parallel Operating Environment (POE) seems to use its MPI implementation efficiently within a node. For this purpose a special environment variable, MP_SHARED_MEMORY, has been introduced by POE to specify the use of shared memory. However, the focus of this paper is on the implementation of domain cloning which took only a couple of days. The implementation of a hybrid MPI/OpenMP paradigm in the TORB code would take significantly longer and does not promise to be a significant improvement for its scaling property.

4. Results

As pointed out in the introduction realistic global nonlinear PIC simulations already consist of >100 million Monte Carlo particles. Such simulations can be only realized on massively parallel computers. Thus, the implementation and testing of the domain cloning concept as part of the TORB code took place at the Computer Center of the Max Planck Society and the Max Planck Institute for Plasma Physics on a maximum number of 512 processors (PEs). However the final results have been achieved on one of the two IBM Cluster 1600 supercomputer systems at the High Performance Computing Facility of the European Centre for Medium-Range Weather Forecast (ECMWF). Each separate cluster comprises 70 pSeries 690+ servers (compute nodes), each of which has 32 1.9 GHz Power 4+ CPUs. Up to 64 compute nodes with 32 GB memory each have been available for the largest simulation shown here.

4.1. Single node efficiency

In Fig. 2(a) the speed-up over the number of PEs is shown for a 50 million particles simulation with domain decomposition on a dedicated compute node. The problem size is kept constant while increasing the number

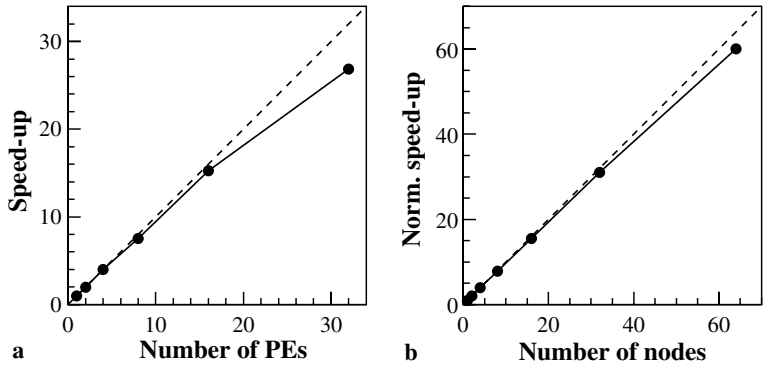


Fig. 2. (a) the speed-up vs. the number of processors (PEs) on a single compute node for a constant problem size. (b) the speed-up with normalized simulation size vs. the number of compute nodes involved. The problem size scales linearly with the number of nodes. The dashed line depicts ideal scaling.

of PEs assigned to the simulation (hard scaling). One can see that the TORB code scales very close to linearly up to 16 PEs. A total speed-up factor of 27 for a compute node consisting of 32 PEs is achieved. This ratio of 27/32 has to be interpreted with care because the single PE run had exclusive usage of the hardware as all the other PEs were idling. This is not consistent with a realistic operation modus of a computer. A more realistic benchmark would have been to perform 32 simulations simultaneously, one on each PE, which was not possible in our case as one simulation already occupies the full memory of the node dedicated to the user. Nevertheless, the scaling shows that a domain decomposition on a single node is reasonable for a number of 32 PEs and results in a total flop rate of 24.3 Gflop/s which is 10% of the theoretical peak performance per node. The flop rate has been measured with a library which API (Application Programming Interface) provides access to the hardware performance counters built-in the IBM Power 4+ CPU.

The benchmark shows a gap between the measured and optimal speed-up of ≈15%. This is basically an effect of the decrease in efficiency when scaling the benchmark from 16 PEs to 32 PEs. It could be well possible that the hybrid MPI/OpenMP paradigm with an exclusive OpenMP communication within the node would further increase the speed-up. But even with the hybrid MPI/OpenMP paradigm, it is not guaranteed to achieve an optimal speed-up as other obstacles as e.g. a too small memory bandwidth could also limit the speed-up.

The physical results show that a maximum of 32 cubic B-splines in the direction of the parallelization is well suited to resolve the evolving turbulent field structures along the magnetic field lines [1]. The presence of a magnetic field in the plasma makes the turbulent structures highly anisotropic, with scales of the order of the physical domain along the field lines and as small as a few ion gyration radii perpendicular to the field lines. Thus, when parallelizing along the field lines, a complete physical domain fits onto one node which in the concept of domain cloning makes it appropriate to put each clone onto a node (see previous section).

4.2. Cluster efficiency

In the second benchmark the size of Monte Carlo particles doubles as the number of nodes involved in the simulation is doubled (weak scaling). Starting with 50 million particles for a single node simulation the number of particles increases linearly with the number of nodes. The maximum number of nodes used is 64 which corresponds to a total number of 3.2 billion particles in the simulation. Such a large number of particles gives the opportunity to simulate large physical domains with a very low level of statistical noise for long simulation intervals. Fig. 2(b) shows the speed-up with normalized simulation size as a function of involved nodes n. The scaling proves to be excellent for up to 32 nodes. Only a further doubling of the number of nodes to $n = 64$ gives a significant decrease of the relative speed-up to 60/64. Thus, the intensive all-reduce communication (global sum) over the clones finally diminishes the very good scaling properties of the domain cloning concept. Nevertheless a total flop rate of 1.46 Tflop/s is reached on 64 nodes (2048 PEs) which is ≈9% of the theoretical peak performance.

5. Summary and conclusions

The additional degree of freedom of the domain cloning concept, which mixes between domain decomposition and particle decomposition has been used to optimize the scaling properties of a PIC code on a cluster of SMP computers. In the simulation presented here a complete cloned physical domain is assigned to each compute node. Thus, the domain decomposition concept is only applied to processors on the same node so that the data intensive particle communication is restricted to the shared memory within nodes. Between the nodes only the moderate field communication takes place via the network (fast interconnect). Hence, one has decoupled the parallelization into two completely independent communication processes and therefore two different scaling laws: one concerning a single node depending on the number of processors (PEs) used on the node and the other one depending only on the number of nodes (not processors) involved in the communication process. The concept of domain cloning fits exactly to the hardware constraints which makes it possible to run particle-in-cell simulations efficiently up to thousands of processors on a cluster of SMP computers. A total sustained performance of 1.5 Tflop/s was achieved for the TORB code on such a machine of 2048 processors.

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